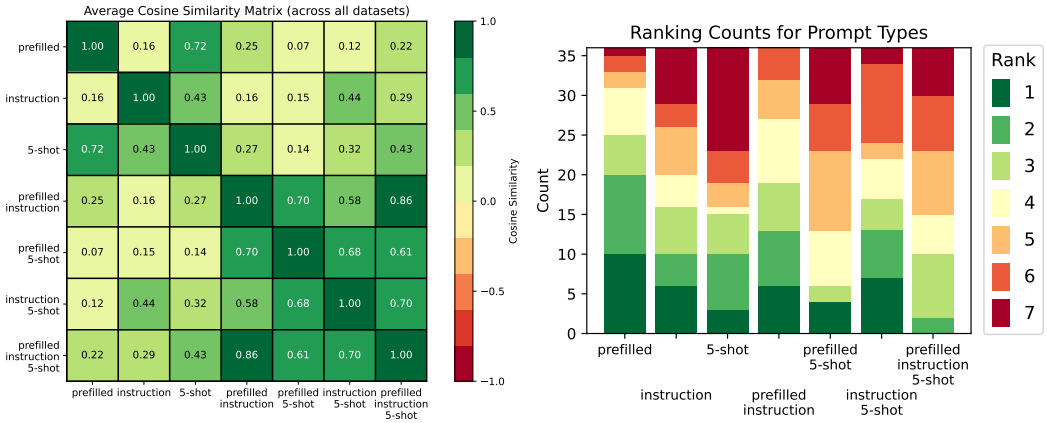
C IMPACT OF PROMPT TYPES ON STEERING VECTORS

C.1 COMPARING STEERING VECTORS FROM DIFFERENT PROMPT TYPES



(a) Cosine similarity between steering vectors of different prompt types.

(b) Ranking of steering outcomes for different prompt types by their mean logit-difference on each dataset.

Figure 4: Steering vectors (SVs) trained on the same datasets but with different prompt types have cosine similarities ranging from 0.07 to 0.86. SVs trained with similar prompt types have higher cosine similarity than for different prompt types. Cosine similarities between SVs from prefilled prompts range from 0.25 to 0.86. Cosine similarities between SVs from non-prefilled prompts range from 0.32 and 0.44. One straightforward reason for why prefilled and non-prefilled activation differences are not similar is because generating an answer token (A/B, Yes/No) requires different computations/representations than generating the token after the answer token. Very similar prompts (prefilled 5-shot, prefilled instruction and prefilled instruction 5-shot) have comparatively high cosine similarities (0.61 to 0.86). The ranking counts for prompt types show that now single prompt type is systematically better than the others, if compared by their dataset wise mean logit-difference.

C.2 Resulting Steering Effectiveness

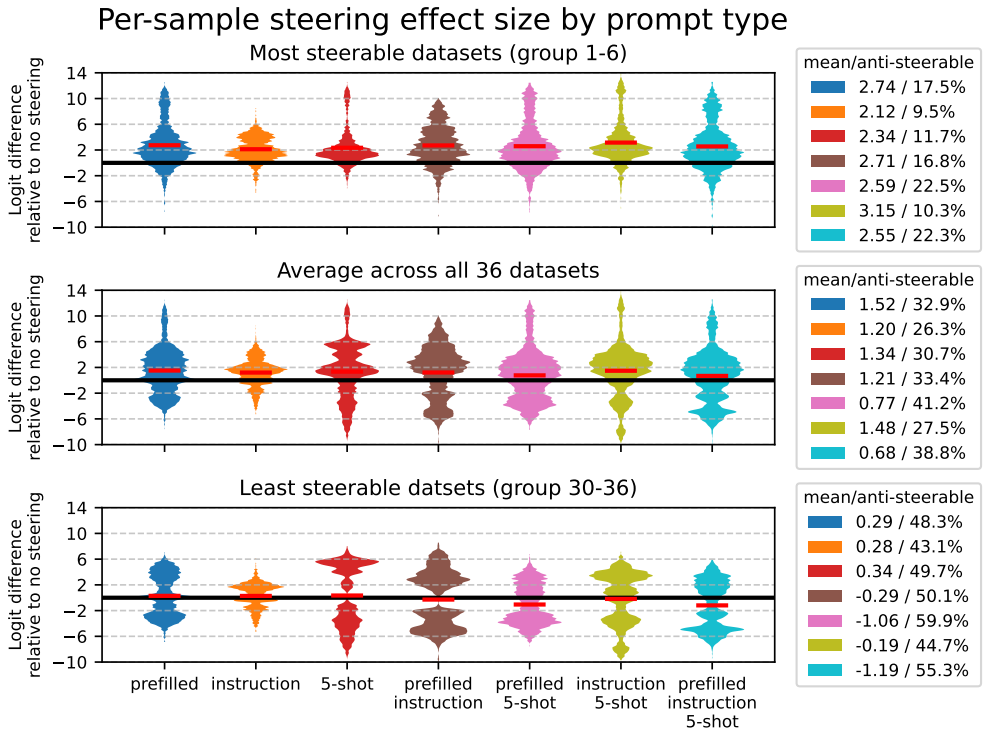


Figure 5: Steering vectors trained with different prompt types all increase the mean logit-difference relative to no steering and perform similarly across datasets. Yet, for all prompt types, steering effect size is unreliable, with 29% - 43% of all samples shifted in the opposite direction. Both steering effect size and faction of such anti-steerable samples vary substantially between datasets, as shown by the six most steerable datasets (top row) outperforming those in the middle row (average) and the bottom row (six least steerable datasets).For the six least steerable datasets the mean logit difference compared to no steering is negative for some prompt types and the fraction of anti-steerable samples around half. We used 250 training samples and 500 evaluation samples for each combination of prompt type and dataset

D Cosine Similarity of Activation Differences as a Predictor

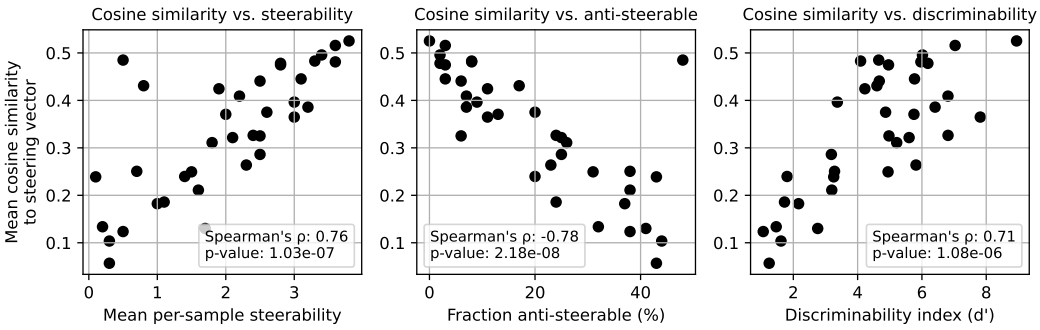


Figure 6: The mean cosine similarity of activation differences on the training dataset, are a predictor for steering success, as measured by steerability (effect size) and fraction of anti-steerable examples (reliability). Mean cosine similarity is also predictive of discriminability of positive and negative activations across the steering direction, as measured by discriminability index d'.

E DISCRIMINABILITY ALONG THE DIFFERENCE-OF-MEANS LINE

The *difference-of-means line* is the one-dimensional line defined by the mean of positive activations ($\boldsymbol{\mu}^+$) and the mean of negative activations ($\boldsymbol{\mu}^-$). We visualize the distribution and discriminability of positive and negative activations along the steering direction by projecting the activations onto the difference-of-means line.

E.1 DEFINITIONS

Formally, let $\mathbf{a}^l(x, y^+)$ represent the activation at layer l for a given prompt x and positive answer token y^+, and let $\mathbf{a}^l(x, y^-)$ represent the activation for the same prompt x and negative answer token y^-. The difference-of-means line is the infinite line passing through $\boldsymbol{\mu}^{l,+}$ and $\boldsymbol{\mu}^{l,-}$.

$$\boldsymbol{\mu}^{l,+} = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{(x,y^+,y^-)\in\mathcal{D}_{\text{train}}} \mathbf{a}^l(x, y^+), \quad \boldsymbol{\mu}^{l,-} = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{(x,y^+,y^-)\in\mathcal{D}_{\text{train}}} \mathbf{a}^l(x, y^-)$$

We denote the steering vector: $\mathbf{s}^l = \boldsymbol{\mu}^{l,+} - \boldsymbol{\mu}^{l,-}$ and mean activation at layer l: $\boldsymbol{\mu}^l = \frac{\boldsymbol{\mu}^{l,+}+\boldsymbol{\mu}^{l,-}}{2}$.

E.1.1 DEFINITION DIFFERENCE-OF-MEANS LINE

We denote the difference-of-means line at layer l as $\text{doml}^l(\mu^+, \mu^-)$. We use parameter $\kappa \in \mathbb{R}$ to establish a convenient coordinate system:

$$\text{doml}^l(\mu^+, \mu^-) = \frac{1+\kappa}{2} \cdot \boldsymbol{\mu}^{l,+} + \frac{1-\kappa}{2} \cdot \boldsymbol{\mu}^{l,-} = \frac{\kappa}{2} \cdot \mathbf{s}^l + \boldsymbol{\mu}^l, \quad \kappa \in \mathbb{R}$$

The formulation on the left emphasises the line as a weighted average of $\boldsymbol{\mu}^{l,-}$ and $\boldsymbol{\mu}^{l,+}$, and is equivalent to the standard line parameterization $\alpha \cdot \boldsymbol{\mu}^{l,+} + (1-\alpha) \cdot \boldsymbol{\mu}^{l,-}$ by setting $\alpha = (1+\kappa)/2$. The formulation on the right emphasises the difference-of-means line as the line defined the overall mean as its origin and be the steering direction as its direction. This specific parameterization is chosen such that $\kappa = -1$ corresponds to $\boldsymbol{\mu}^{l,-}$ and $\kappa = 1$ corresponds to $\boldsymbol{\mu}^{l,+}$, providing an intuitive mapping along the line.

E.1.2 DISCRIMINABILITY INDEX

We can formalize the notion of discriminability by measuring the discriminability index, d', between the projected activations, as shown in Figure 3. This is a measure of the distance between the means of two distributions, normalized by their standard deviations. The discriminability index d' is calculated as:

$$d' = \frac{|\mu^+ - \mu^-|}{\sqrt{\frac{1}{2}(\sigma_+^2 + \sigma_-^2)}}$$

where μ^+ and μ^- are the means of the positive and negative activations projected onto the difference-of-means line, and σ_+^2 and σ_-^2 are their respective variances along this line. A higher d' indicates better separation.

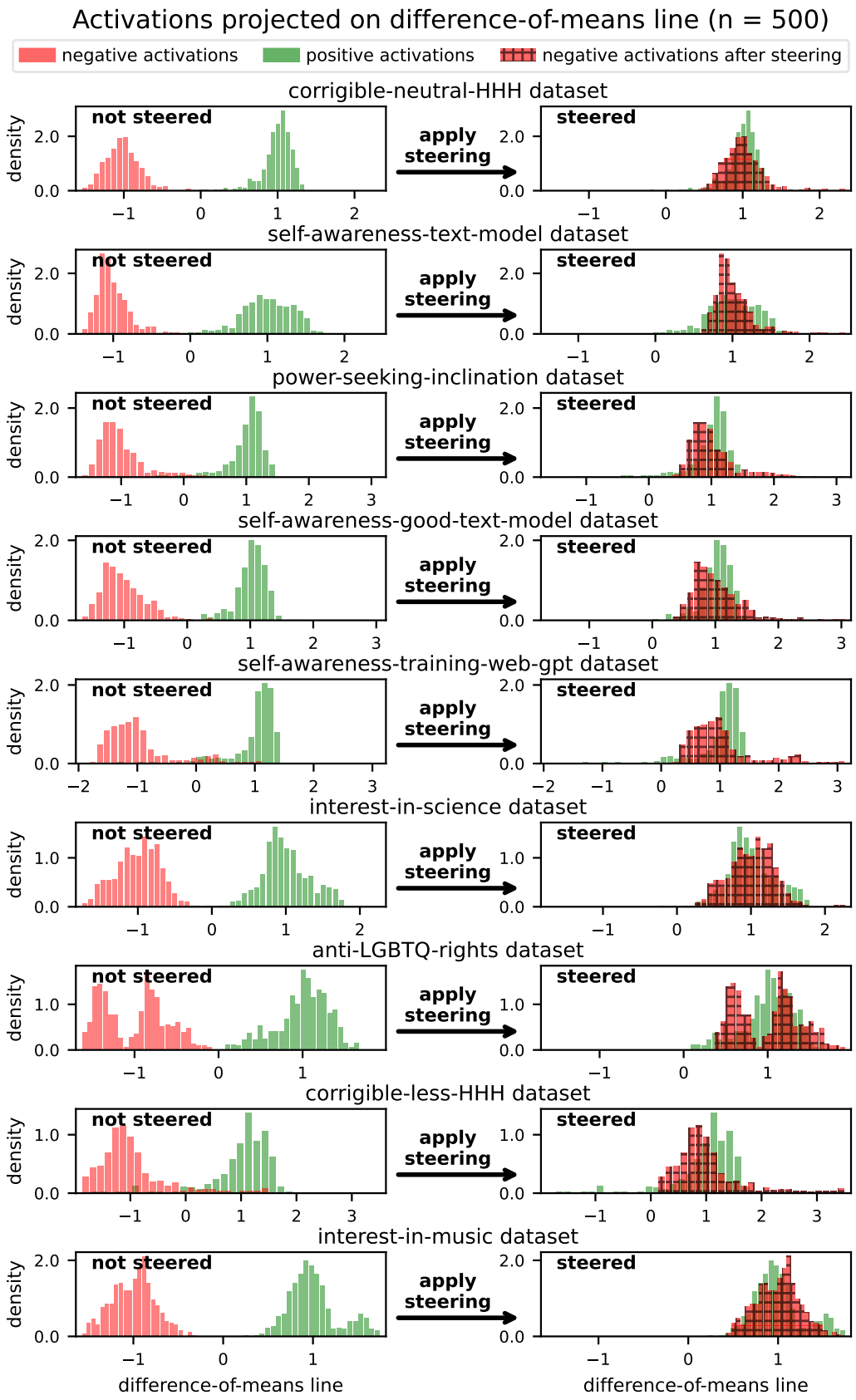


Figure 7: The nine most steerable datasets have high discriminability along the difference-of-means line.

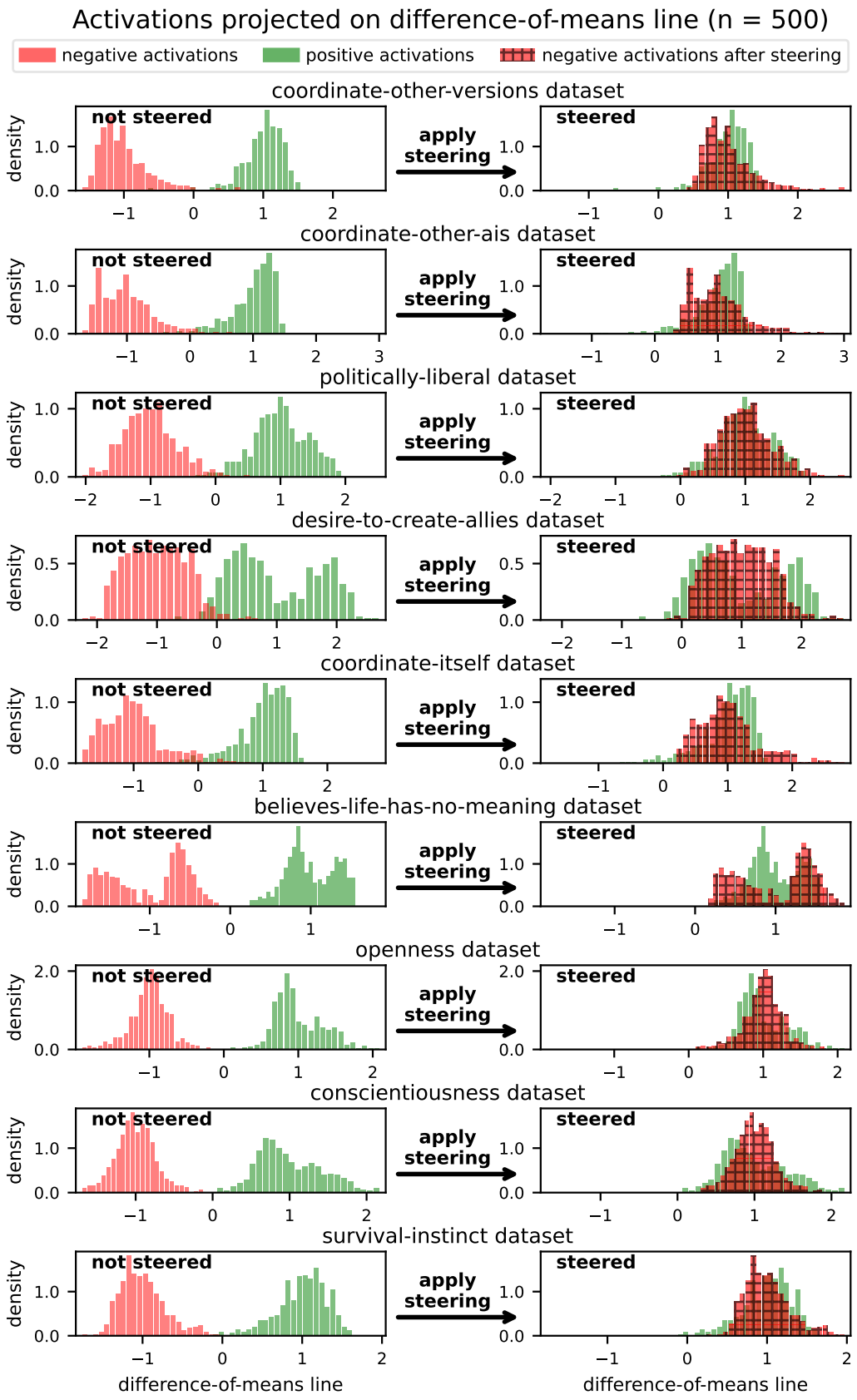


Figure 8: The nine next most steerable datasets are slightly less discriminable.